

Semester	Course Code	Course Category	Hours/ Week	Credits	Marks for Evaluation		
					CIA	ESE	Total
I	23UMA1AC1	Allied - I	4	3	25	75	100
Course Title Linear Algebra and Differential Equations							

SYLLABUS		
Unit	Contents	Hours
I	Matrices- *Special types of matrices*- Scalar multiplication of a matrix- Equality of matrices, Addition of matrices- Subtraction- Symmetric matrix-Skew symmetric matrix-Hermitian and skew Hermitian matrices- Multiplication of matrices- Inverse matrix- Orthogonal matrices (Problems only).	12
II	Solution of simultaneous equations-Rank of a matrix- Eigen values and Eigen vectors-*Cayley Hamilton theorem*. (Problems only)	12
III	Differential equations of the first order with higher degree – Equations solvable for p- Equations Solvable for y – *Equations Solvable for x* - Clairaut's form. (Problems only)	12
IV	*Linear Differential Equations with constant coefficients * - Particular integral – Special method of finding P.I. – Derivation of partial differential equations by elimination of arbitrary constants and arbitrary functions – Different integrals of First Order P.D.E. (Problems only)	12
V	Standard type of first order partial differential equations I, II, III and IV (Clairaut's form) - *Lagrange's equations*. (Problems only).	12

..... Self Study

Text Books:		
1. T.K. Manicavachagom Pillay, T. Natarajan and K.S. Ganapathy, Algebra Volume-II, Ananda Book Depot, Chennai (2019) 2. S. Narayanan, T.K. Manicavachagom Pillay, Calculus Volume-III, S. Viswanathan Publishers Pvt. Ltd. (2012).		
UNIT I	Chapter 2 Sections 1-9	T.B-1
UNIT II	Chapter 2 Sections 10-13, 16	T.B-1
UNIT III	Chapter 1 Sections 5.1–5.4, 6.1, 6.2	T.B-2
UNIT IV	Chapter 2 Sections 1–4 Chapter 3 Sections 1–3	T.B-2
UNIT V	Chapter 4 Sections 5.1-5.4, 6	T.B- 2
Reference Books:		
1. P. Kandasamy and K. Thilagavathy, Allied Mathematics, S. Chand & Company Ltd, New Delhi (2010). 2. A. Abdul Rasheed, Allied Mathematics, Vijay Nicole Imprints private limited, Chennai (2008). 3. S. Arumugam and A. Thangapandi Isaac, Ancillary Mathematics, New Gamma Publishing house (2002).		
Web Resources:		
1.	https://nptel.ac.in/courses/111/107/111107111/	
2.	https://nptel.ac.in/courses/111/102/111102133/	

Course Outcomes		
Upon successful completion of this course, the student will be able to:		
CO No.	CO Statement	Cognitive Level (K-Level)
CO1	recognize and recall the basic concept of matrices and first order differential equations with examples.	K1
CO2	compute the operations on matrices and solving differential equations related problems.	K2
CO3	apply the concepts of matrices for solving system of equations, Eigen values and Eigen vectors.	K3
CO4	analyse the impact of an applications of mathematical concepts in computer science using matrices and differential equations.	K4
CO5	evaluate the general solution of ordinary and partial differential equations	K5

Relationship Matrix:

Course Outcomes (COs)	Programme Outcomes (POs)					Programme Specific Outcomes (PSOs)					Mean Score of COs
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3	PSO4	PSO5	
CO1	3	2	3	3	1	3	3	2	3	2	2.5
CO2	3	3	2	2	1	2	3	2	3	2	2.3
CO3	3	3	2	2	1	3	2	3	3	2	2.4
CO4	3	2	3	3	2	3	3	3	2	2	2.6
CO5	2	2	2	3	1	2	2	2	3	2	2.1
Mean Overall Score											2.38
Correlation											Medium

Mean Overall Score	Correlation
< 1.5	Low
≥ 1.5 and < 2.5	Medium
≥ 2.5	High

Course Coordinators:

Dr. M.A. Rifayathali

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